

- Q.26 How does emf induced in a coil rating in magnetic field?
- Q.27 What is the reason behind the lamination of stator in an induction motor?
- Q.28 Explain the constructional details of DC generator.
- Q.29 What is the use of potential transformer? Explain its working with constructional details.
- Q.30 How commercial efficiency is different from all day efficiency of a transformer?
- Q.31 What are the application of synchronous machine?
- Q.32 Write a short note on commutator type single phase motor.
- Q.33 Write a short note on micromotors.
- Q.34 What is the need of damper winding in synchronous motor?
- Q.35 Write down the difference between servo motors and stepper motors?

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Derive the relationship between phase voltage and line voltage in a delta configuration of three phase supply.
- Q.37 Explain the constructional details of different types of single-phase induction motor.
- Q.38 What is DC machine? What are the factors determining the speed of DC motor? What are the applications of DC machine?

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3rd Sem / Eltx, EI, IC, Medical Eltx. (5th Sem.)

Power Eltx, Elect. & Eltx. Engg.

Sub : Electrical Machines

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Which of the following is the primary function of a commutator in a DC machine?
- To reverse the current direction
 - To maintain constant field flux
 - To regulate the armature speed
 - To reduce the power loss
- Q.2 In an induction motor, which of the following factors determines the speed of the motor?
- Frequency of the supply
 - Resistance of the rotor
 - Number of poles in the motor
 - Voltage applied to the stator
- Q.3 What are the two main types of servo motor?
- AC and DC
 - Stepper and brushless
 - Permanent magnet and variable reluctance
 - Linear and rotary
- Q.4 Which is the most widely used material in the core of the transformer?
- Cold rolled grain-oriented sheet steel
 - Cold rolled grain steel

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- c) Soft iron
- d) Steel
- Q.5 CVT stands for?
 - a) Constant voltage transducer
 - b) Current voltage transformer
 - c) Cathode voltage transformer
 - d) Constant voltage transformer
- Q.6 Rotor rotates in _____ direction in a three-phase induction motor?
 - a) The clockwise direction only
 - b) The direction of RMF
 - c) The anticlockwise direction only
 - d) Depends on Load
- Q.7 ADC series motor is best suited for electric traction:
 - a) Lathe b) Machine Tool
 - c) Crane d) Constant speed load
- Q.8 In a DC generator, what determines the direction of induced EMF?
 - a) Direction of the armature current
 - b) Direction of the magnetic field and motion
 - c) Speed of rotation of the armature
 - d) Resistance of the winding
- Q.9 Which of the following is the primary advantage of 3-phase induction motor?
 - a) Higher starting torque
 - b) Smooth operation with less vibration
 - c) High efficiency
 - d) Simple construction and low cost
- Q.10 _____ type of coils are used for winding the single phase induction motor generally?
 - a) Rectangular coils b) Square coils
 - c) Cruciform coils d) Circular coils

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SECTION-B

- Note:** Objective type questions. All questions are compulsory. (10x1=10)
- Q.11 Define single phase motor.
 - Q.12 Power factor is the measure of how effectively electrical power is converted into useful work output in an AC circuit. (True / False)
 - Q.13 Write one application of instrument transformer.
 - Q.14 The phase voltage and line voltage are the same for _____ connection.
 - Q.15 What is magnetic flux?
 - Q.16 Define EMF in a coil.
 - Q.17 What is CTs?
 - Q.18 Write one application of synchronous machine.
 - Q.19 Give one application of DC machine.
 - Q.20 How do we can control speed of a DC servo motor?

SECTION-C

- Note:** Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)
- Q.21 Write three applications of both servo motor and stepper motor.
 - Q.22 Define generator and motor. Write down the basic principle of it.
 - Q.23 Define voltage regulation in transformer. What are the losses in transformer?
 - Q.24 State the Faraday's law of electromagnetic induction.
 - Q.25 What is torque and how it is produced with the help of two magnetic fields?

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